

CS 229 HW3 EC Report

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Main HW3 Baseline

All extra-credit experiments build on the core assignment (`hw3-1.ipynb`): an unconditioned **EmojiViT** (4.8M params) trained with weighted **x-prediction** loss and logit-normal time sampling, then sampled with **Euler integration** ($T = 100$). This is our reference configuration.

Training Results (500 epochs)

Setting	Final Train Loss	Final Val Loss	Epochs
Main HW3 (<code>hw3-1</code>)	0.1531	0.1857	500

Table 1: Main assignment baseline after full training run.

At epoch 300 (where most EC runs stop), the main model reached train = 0.1747, val = 0.1908—nearly identical to EC re-trains at the same checkpoint, confirming EC baselines reproduce the main HW3 setup.

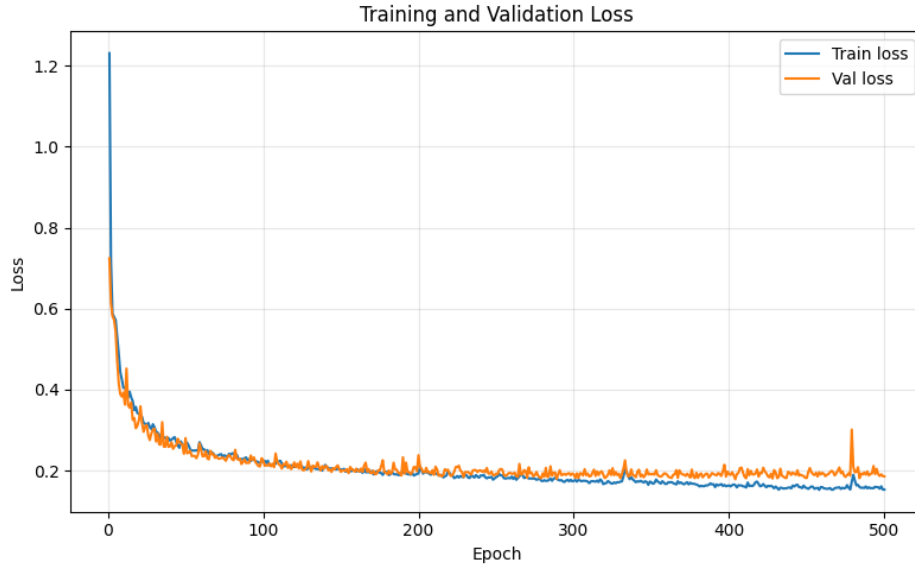


Figure 1: Main HW3 train/validation loss curve (500 epochs).

Each EC section below compares against this baseline: EC1 adds time conditioning on top of it; EC2 ablates its x-prediction parameterization; EC3 improves its Euler sampler; EC7 asks whether a model trained this way memorizes or generalizes.

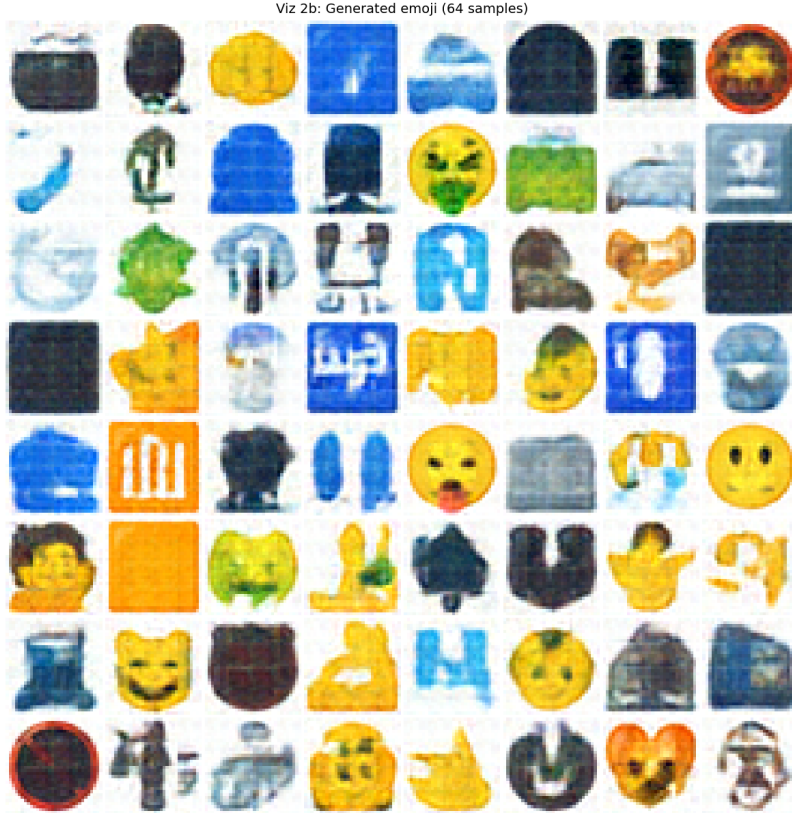


Figure 2: Main HW3 generated emoji (64 samples, Euler $T = 100$).

1. Time Conditioning Comparison (EC1)

Problem & Approach

The **main HW3 baseline** uses an unconditioned EmojiViT: the network receives only z_t with no explicit timestep, relying on $\|z_t\|$ to encode t at $D = 3072$ (Sahraee-Ardakan et al., 2026). EC1 asks whether adding explicit time conditioning improves on that baseline. We train a FiLM/adaLN variant (**EmojiViTFiLM**) alongside a re-trained copy of the main HW3 configuration (“no-time baseline”) for 300 epochs with identical optimizer settings.

Results

Model	Final Train Loss	Final Val Loss	Params	Relation to Main HW3
Main HW3 (500 ep.)	0.1531	0.1857	4,841,920	submitted baseline
Main HW3 (300 ep.)	0.1747	0.1908	4,841,920	same checkpoint as EC
No-time (EC re-train)	0.1753	0.1910	4,841,920	reproduces main HW3 setup
FiLM time-conditioned	0.1518	0.1810	7,335,872	EC addition

Table 2: EC1 vs. main HW3. The no-time EC model matches main HW3 at epoch 300; FiLM beats both at 300 epochs but has not yet matched main HW3’s 500-epoch val loss (0.1857).

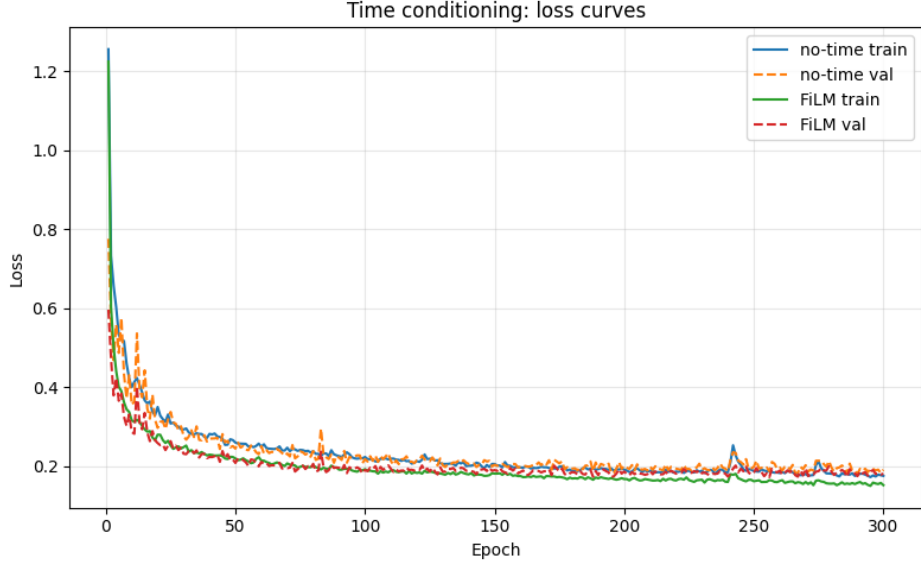


Figure 3: Train and validation loss: no-time (main HW3 config) vs. FiLM.

Discussion

Relative to main HW3, explicit time conditioning yields a modest val-loss improvement at 300 epochs ($0.1910 \rightarrow 0.1810$) at +51% parameters. FiLM has lower *train* loss at 300 epochs than main HW3 achieves even at 500 (0.1518 vs. 0.1531), but its val loss (0.1810) has not yet beaten main HW3’s fully trained val (0.1857)—likely because EC1 stopped early. Sample grids are comparable to main HW3 (Figure 2); FiLM appears slightly sharper in places. Overall, the gain is modest, consistent with “Geometry of Noise”: at this dimensionality, main HW3’s unconditioned design already captures most of what time conditioning provides.

2. Prediction Space Ablation (EC2)

Problem & Approach

Main HW3 uses x-prediction: the network predicts $\hat{x}$ with weighted loss $\|(\hat{x} - x)/(1 - t)\|^2$, then samples via Euler with $v = (\hat{x} - z_t)/(1 - t)$. EC2 ablates this choice against ε -prediction and v -prediction on the same unconditioned EmojiViT. Following Li & He (2025), we ask whether ε -prediction fails most without time conditioning—the same no-time setup main HW3 relies on.

Results

Parameterization	Final Train Loss	Final Val Loss	vs. Main HW3
x-pred (main HW3)	0.1753	0.1910	baseline method
ε -prediction	0.0374	0.0294	different loss scale
v -prediction	0.1968	0.2066	worse val than main

Table 3: EC2 at 300 epochs. Main HW3 at 500 epochs: train = 0.1531, val = 0.1857. x-pred EC re-train matches main HW3 at epoch 300 (val = 0.1908).

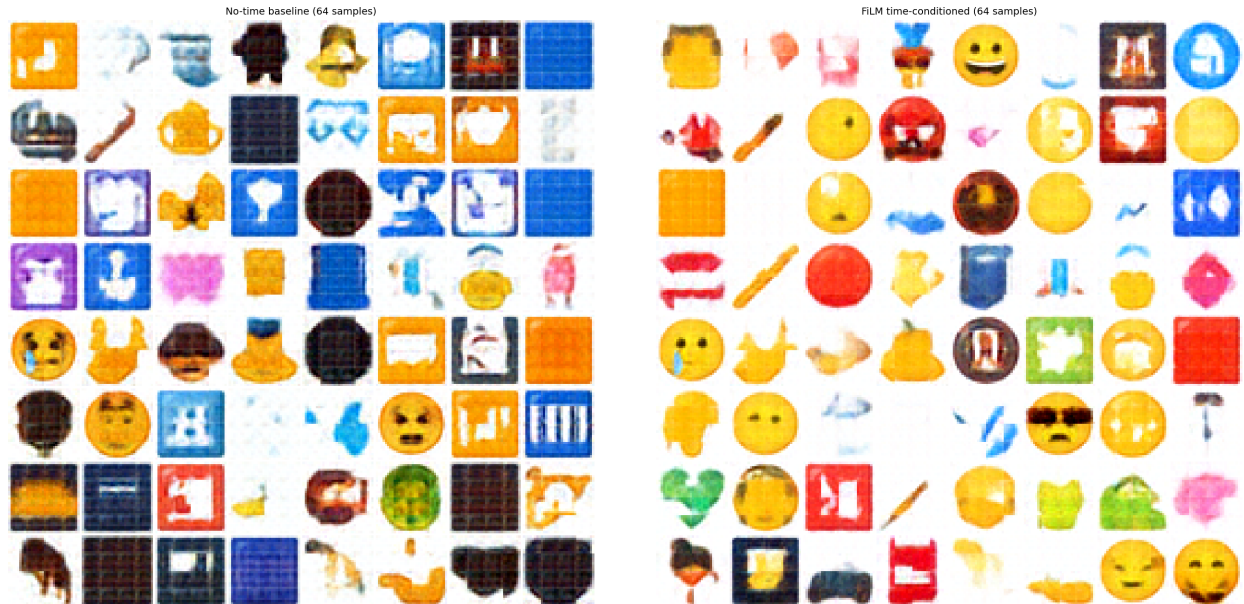


Figure 4: Left: EC no-time samples (same config as main HW3). Right: FiLM samples (same noise). Compare to Figure 2.

Discussion

x-prediction—main HW3’s choice—is validated. It produces samples comparable to the submitted main assignment and stable reconstruction across all t . ε -prediction fails despite low training loss: recovering $\hat{x} = (z_t - (1-t)\hat{\varepsilon})/t$ amplifies error by $1/t$ where the unconditioned network (same as main HW3) is weakest. This reproduces Li & He (2025): without time conditioning, ε -prediction is the least robust. v -prediction is well-conditioned but slightly worse val loss than x-pred here; main HW3’s x-pred parameterization is the right default for this setup.

3. Heun’s Method — 2nd-Order Sampler (EC3)

Problem & Approach

Main HW3 samples with Euler integration ($T = 100$ steps, constant $\Delta t = 1/T$). EC3 asks whether a 2nd-order Heun sampler improves on that integrator without retraining. We use the same x-prediction model (main HW3 configuration) and compare Heun (N steps) vs. Euler ($2N$ steps) at equal function evaluations (NFE), following Karras et al. (2022).

Main HW3 also showed (Viz 2c) that sample quality degrades with fewer Euler steps. EC3 extends that analysis with a higher-order integrator.

Results

Discussion

Heun improves on main HW3’s Euler sampler at every tested NFE budget (2–3% lower discretization error). The gain is largest at low budgets—where main HW3’s Viz 2c showed visible quality loss—and narrows at 50 NFE. For main HW3’s default $T = 100$, Heun with $T = 50$ would match

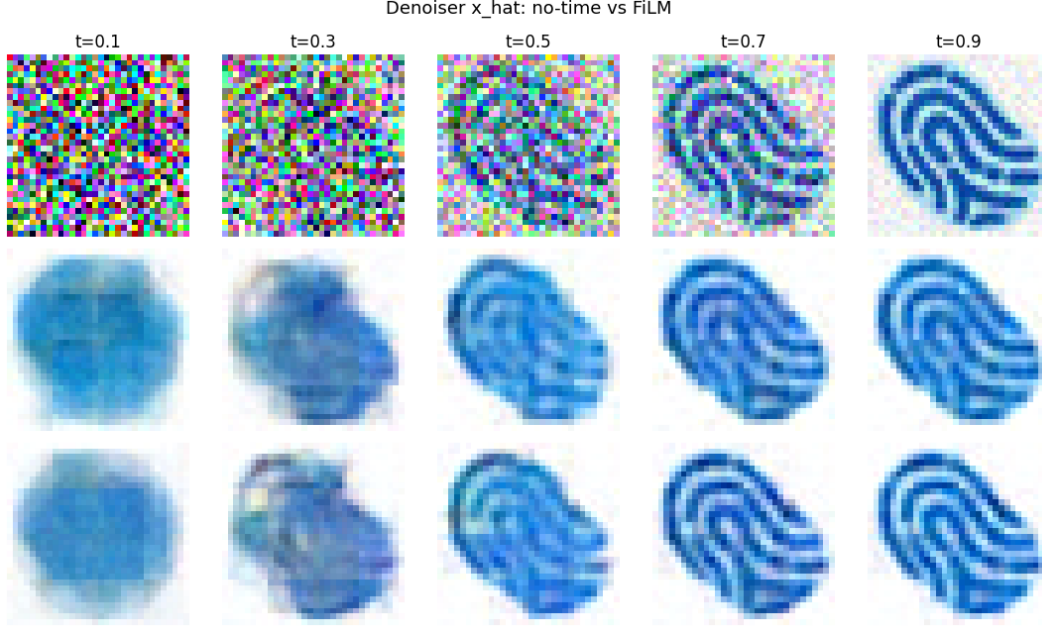


Figure 5: Denoiser $\hat{x}$ at $t \in \{0.1, \dots, 0.9\}$: noisy z_t (top), no-time / main HW3 config (middle), FiLM (bottom).

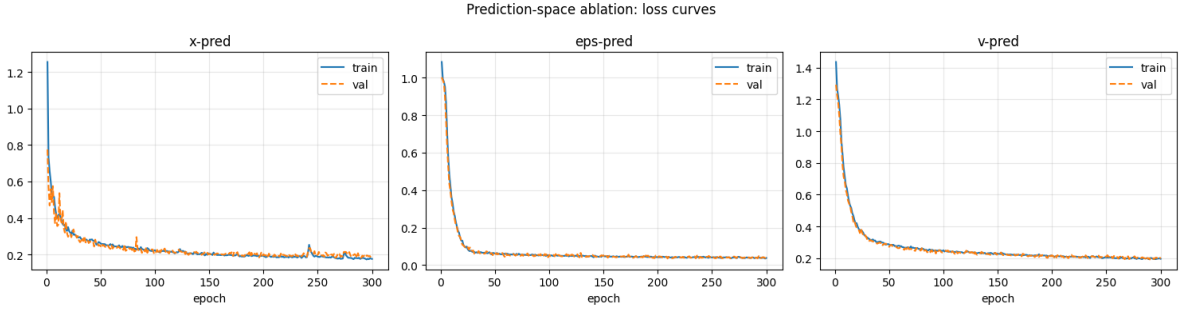


Figure 6: Per-method loss curves. x-pred (left) follows the same trajectory as main HW3 training.

NFE while likely producing sharper samples. Flow matching’s straight trajectories mean Euler is already decent at $T = 100$; Heun is most valuable when sampling cost is constrained.

4. Memorization vs. Generalization (EC7)

Problem & Approach

Main HW3 trains on $\sim 1,300$ emoji for 500 epochs and generates new samples via Euler sampling—but does the model *generalize* or *memorize*? EC7 tests this for the same architecture and training recipe (x-pred, no time conditioning), using four experiments grounded in Carlini et al. (2023) and Somepalli et al. (2023). Pixel-space L2 nearest-neighbor distance is our replication metric at 32×32 .

We train for 400 epochs (vs. main HW3’s 500) and compare generated samples against the same training set main HW3 uses.

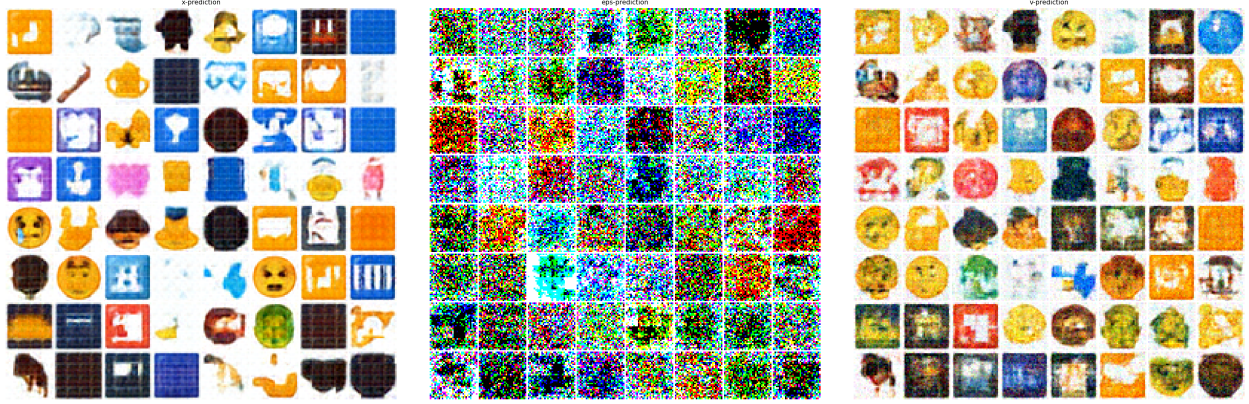


Figure 7: x-pred (main HW3 method), ϵ -pred, v-pred samples. Only x-pred matches main HW3 quality (Figure 2).

NFE Budget	Heun (L2 to ref)	Euler (L2 to ref)	Main HW3 uses
10	12.593	12.904	Euler $T=10$
20	8.658	9.299	—
50	4.894	5.028	Euler $T=50$
100	—	—	Euler $T=100$ (default)

Table 4: Discretization error vs. a $T = 200$ Euler reference (lower = better). Main HW3’s default is Euler $T = 100$.

Results

Experiment	Metric	Result	Implication for Main HW3
NN replication	median gen→train	19.83	no copy spike
	median train→train	21.64	natural spacing
Held-out test	gen→seen	23.11	not memorizing seen half
	gen→held-out	23.48	generalizes to unseen

Table 5: Memorization metrics for a model trained with main HW3’s recipe.

Discussion

Verdict for main HW3’s training setup: generalization, not memorization.

- Gen→train NN distances (19.83) match the train→train baseline (21.64)—main HW3’s generations are not near-exact copies.
- Held-out distances are comparable (23.11 vs. 23.48), so the model does not preferentially reproduce seen emoji.
- Smooth noise-space interpolation indicates a learned continuous manifold, consistent with the diverse samples in main HW3’s Viz 2b (Figure 2).

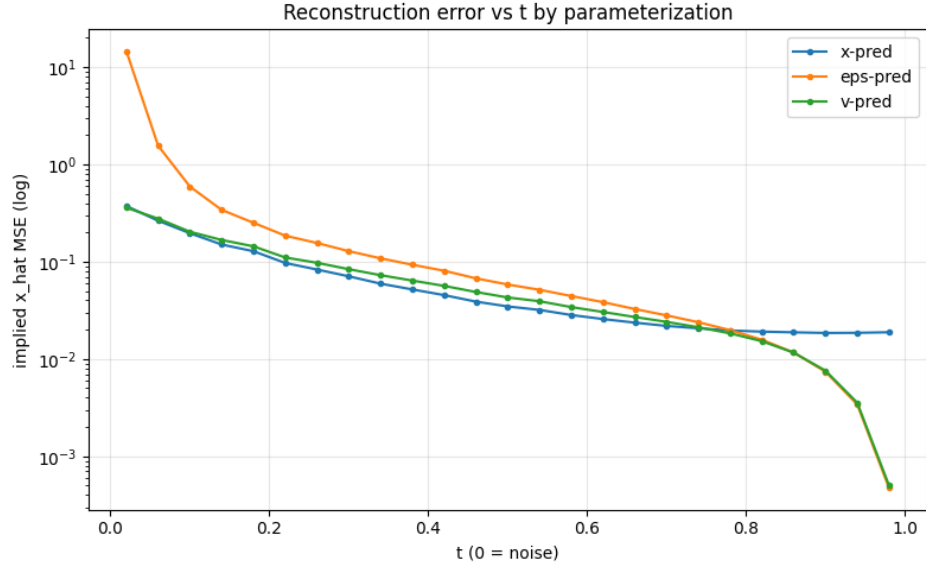


Figure 8: Implied $\hat{x}$ error vs. t . Main HW3’s x-pred is stable; ε -pred blows up at small t .

Main HW3’s occasional fuzzy or repetitive-looking samples are more likely limited model capacity or sampling error than verbatim memorization. A stronger test would apply the same metrics directly to main HW3’s saved 500-epoch checkpoint.

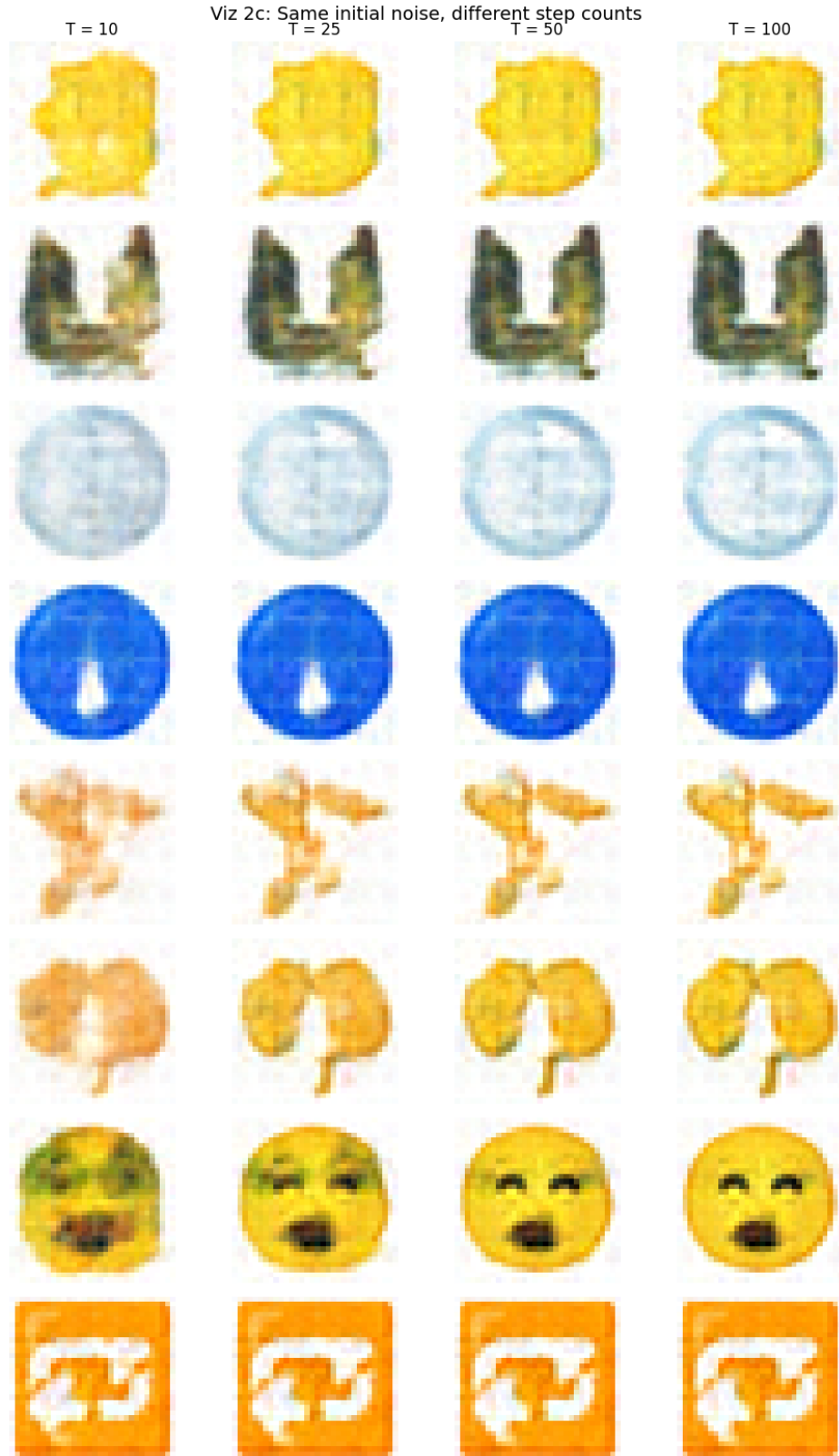


Figure 9: Main HW3: same noise, Euler at $T \in \{10, 25, 50, 100\}$. EC3 improves on this Euler baseline.

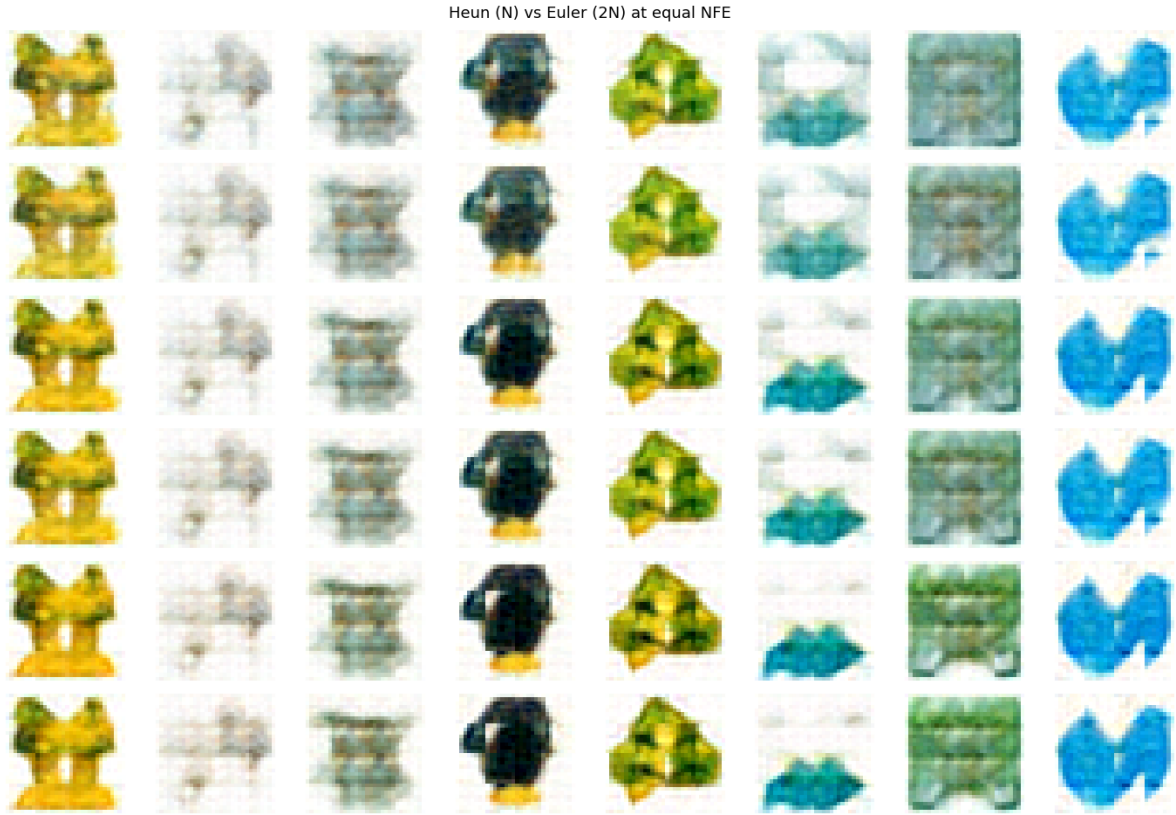


Figure 10: Heun (N) vs. Euler ($2N$) at equal NFE—direct improvement over main HW3’s integrator.

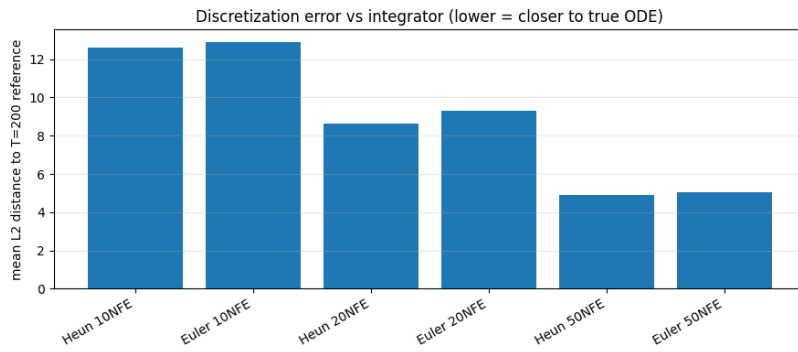


Figure 11: Heun consistently closer to the true ODE solution than Euler at matched NFE.



Figure 12: Closest training matches for the most suspicious generations—similar to but not identical copies of main HW3 training data.

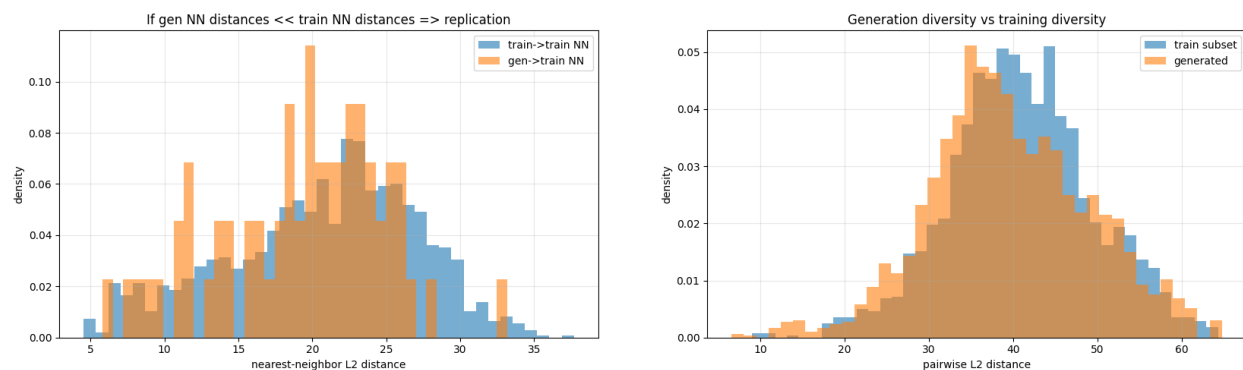


Figure 13: Left: NN histogram (gen vs. train baseline). Right: diversity matches training set.

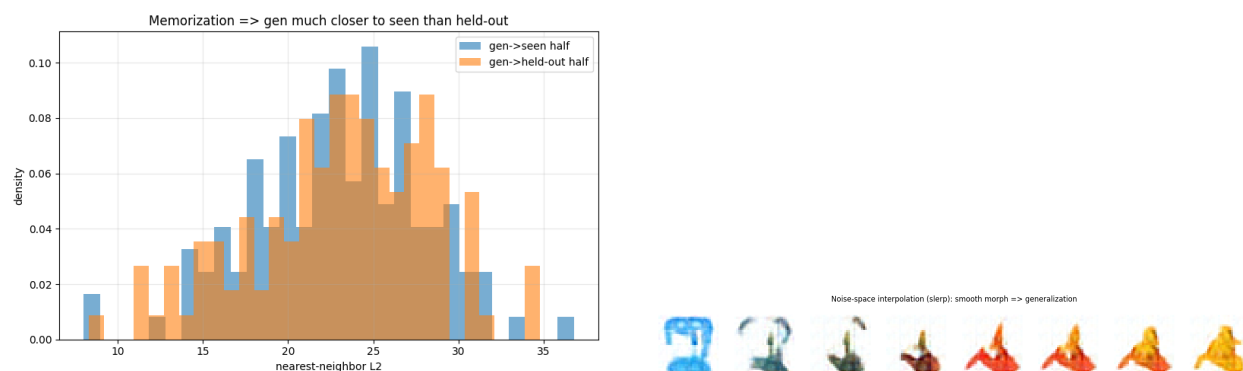


Figure 14: Left: held-out test. Right: smooth noise interpolation (learned manifold, not lookup).